University of Ottawa

Department of Mathematics and Statistics

MAT 1302C : Mathematical Methods II

Professor: Hadi Salmasian

Final Exam

December 8, 2012

Surname _____ First Name _____

Student # _____

Instructions:

(a) You have 3 hours to complete this exam.

(b) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(c) Write your student number at the top of each page in the space provided.

(d) No notes, books, scrap paper, calculators or other electronic devices are allowed.

(e) You are strongly recommended to write in **pen**, not pencil.

(f) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1–13	14	15	16	17	18	19	20	21	Total
Maximum	26	5	3	5	6	6	4	6	4	65
Grade										

Part A: Answer Only Questions

In Questions 1–13, only your final answer will be considered for marks. Each question is worth 2 points.

1. Let $A = \begin{bmatrix} 1 & 0 & -2 \\ 1 & -\frac{1}{2} & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 3 \\ 5 & 1 \end{bmatrix}$. Compute $A^T B$.

Solution: $A^T B = \begin{bmatrix} 4 & 4 \\ -\frac{5}{2} & -\frac{1}{2} \\ 7 & -5 \end{bmatrix}$

2. Write the complex number $\dfrac{2i-3}{i+3}$ in the form $a+bi$ where a and b are real numbers.

Solution: $-\dfrac{7}{10} + \dfrac{9}{10}i$

3. Let $z = 1+2i$ and $w = i-3$. Compute $z\overline{w}$ and $2z+\overline{w}$.

Solution: $z\overline{w} = -1-7i$ and $2z+\overline{w} = -1+3i$.

4. Let $A = \begin{bmatrix} -3 & 5 & -4 \\ 0 & 2 & -1 \\ 0 & 0 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 0 & 0 \\ -1 & 2 & 0 \\ 6 & 4 & -1 \end{bmatrix}$ and $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Which of the following matrices are invertible?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) ABA^T.
(b) $A - 2I_3$.
(c) $A - B^T$.
(d) $B + AB$.
(e) $B^2 - I_3$.

Solution: (a) and (d).

5. Let $A = \begin{bmatrix} 1 & 0 & -2 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$. What are the dimensions of $\operatorname{Col} A$ and $\operatorname{Nul} A$?

Solution: $\dim \operatorname{Col} A = 3$ and $\dim \operatorname{Nul} A = 2$.

6. Let $A = \begin{bmatrix} 1+i & -i & 2-1 \\ 0 & 1+i & \frac{1}{2}+i \\ 0 & 0 & 0 \end{bmatrix}$. Write down the eigenvalues of A and their multiplicities.

Solution: The eigenvalues are $1+i$ with multiplicity 2 and 0 with multiplicity 1.

7. Let A be a 3×3 matrix with characteristic equation $-\lambda^3 - \lambda^2 + 2\lambda = 0$. Which of the following statements are true?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) A is invertible.
(b) A is diagonalizable.
(c) $\lambda = 2$ is an eigenvalue of A.
(d) $\lambda = 1$ is an eigenvalue of A with multiplicity 2.
(e) The eigenvalues of A^2 are $0, 1$, and 4.

Solution: (b) and (e).

8. Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$ and $B = \begin{bmatrix} 3a_{11} & 3a_{12} & 3a_{13} & 3a_{14} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{41} - 2a_{11} & a_{42} - 2a_{12} & a_{43} - 2a_{13} & a_{44} - 2a_{14} \end{bmatrix}$.

If $\det(A) = 5$ then what is the value of $\det(B)$?

Solution: $\det(B) = (-3) \times 5 = -15$.

9. Let A be a 12×9 matrix (i.e., A has 12 rows and 9 columns) and $\mathbf{b}$ be a vector in $\mathbb{R}^{12}$. We know that the matrix equation $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions. Which of the following statements are **false**?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) $\mathbf{b}$ can be written as a linear combination of the columns of A.
(b) Every row of A has a pivot position.
(c) The columns of A are linearly dependent.
(d) $\operatorname{rank} A \geq 9$.

Solution: (b), (d)

10. Which of the following sets are subspaces of $\mathbb{R}^3$?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) $H = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \;\middle|\; \begin{bmatrix} -1 & 2 & 3 \\ 4 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4x_1 \\ 4x_2 \\ 4x_3 \end{bmatrix} \right\}$

(b) $H = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \;\middle|\; x + y + z \geq 0 \right\}$

(c) $H = \text{Span} \left\{ \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ -6 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix} \begin{bmatrix} -1 \\ -3 \\ 4 \end{bmatrix} \right\}$

(d) $H = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\}$

(e) $H = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \;\middle|\; 2x_1 - 3x_2 + x_3 = 0 \text{ and } x_1 - 4x_3 = 1 \right\}$

Solution: (a), (c).

11. For which pair of values of the parameters p and q does the linear system

$$\begin{cases} -3x_1 + 2x_2 - 5x_3 = -1 \\ x_2 + x_3 = -2 \\ px_2 - qx_3 = p + 3 \end{cases}$$

have infinitely many solutions?

Exactly one of the answers is correct.

(a) $p = 0$ and $q = -3$.
(b) $p = 0$ and $q = 0$.
(c) $p = -2$ and $q = 2$.
(d) $p = -1$ and $q = 1$.
(e) $p = 1$ and $q = 0$.

Solution: (d).

12. Let A be an $n \times n$ matrix. Which of the following statements imply that $\text{rank}\, A = n$?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) $\det A \neq 0$.
(b) The linear system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.
(c) The columns of A are linearly dependent.
(d) Every row of A contains a pivot position.
(e) A^T is invertible.
(f) $\text{Nul}\, A = \mathbb{R}^n$.

Solution: (a), (d), (e).

13. Let $A = PBP^{-1}$ where $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 0 & i \end{bmatrix}$. Compute A^6.

Solution: $\begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix}$

Part B: Long Answer Questions

In Questions 14–21, you must show your work and justify your answers to receive full marks. Partial marks may be awarded for making sufficient progress towards a solution.

14. **(5 points)** Is the following linear system consistent or inconsistent? If it is consistent then find its general solution and write it in vector parametric form.

$$\begin{cases} 3x_1 + 3x_2 + 6x_6 &= 2x_4 + 3 \\ x_2 + 2x_3 + 2x_5 &= x_4 + x_6 \\ 3x_1 + 4x_2 + 2x_3 + 3x_5 + 4x_6 &= 3x_4 + 4 \end{cases}$$

Solution: We find the reduced echelon form of the augmented matrix:

$$\left[\begin{array}{cccccc|c} 3 & 3 & 0 & -2 & 0 & 6 & 3 \\ 0 & 1 & 2 & -1 & 2 & -1 & 0 \\ 3 & 4 & 2 & -3 & 3 & 4 & 4 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_1} \left[\begin{array}{cccccc|c} 3 & 3 & 0 & -2 & 0 & 6 & 3 \\ 0 & 1 & 2 & -1 & 2 & -1 & 0 \\ 0 & 1 & 2 & -1 & 3 & -2 & 1 \end{array}\right]$$

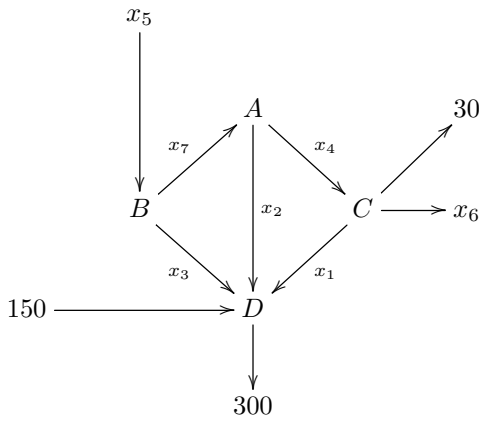
$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{cccccc|c} 3 & 3 & 0 & -2 & 0 & 6 & 3 \\ 0 & 1 & 2 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_3} \left[\begin{array}{cccccc|c} 3 & 3 & 0 & -2 & 0 & 6 & 3 \\ 0 & 1 & 2 & -1 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 3R_2} \left[\begin{array}{cccccc|c} 3 & 0 & -6 & 1 & 0 & 3 & 9 \\ 0 & 1 & 2 & -1 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{array}\right] \xrightarrow{R_1 \to \frac{1}{3}R_1} \left[\begin{array}{cccccc|c} 1 & 0 & -2 & \frac{1}{3} & 0 & 1 & 3 \\ 0 & 1 & 2 & -1 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{array}\right]$$

$$\Rightarrow \begin{cases} x_1 = 2x_3 - \frac{1}{3}x_4 - x_6 + 3 \\ x_2 = -2x_3 + x_4 - x_6 - 2 \\ x_3 : \text{free} \\ x_4 : \text{free} \\ x_5 = x_6 + 1 \\ x_6 : \text{free} \end{cases}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 2x_3 - \frac{1}{3}x_4 - x_6 + 3 \\ -2x_3 + x_4 - x_6 - 2 \\ x_3 \\ x_4 \\ x_6 + 1 \\ x_6 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -\frac{1}{3} \\ 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_6 \begin{bmatrix} -1 \\ -1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ -2 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

15. **(3 points)** The traffic flow in a city is represented by the diagram below. The arrows indicate the directions of one-way traffic. Numbers and variables indicate the number of cars per hour in the corresponding direction. Write down a linear system describing the traffic flow. Do not solve the linear system.



Solution:

$A:\ x_7 = x_2 + x_4$

$B:\ x_5 = x_3 + x_7$

$C:\ x_4 = x_1 + x_6 + 30$

$D:\ 150 + x_1 + x_2 + x_3 = 300 \quad (\text{or } x_1 + x_2 + x_3 = 150)$

$Total:\ x_5 + 150 = 300 + 30 + x_6 \quad (\text{or } x_5 = 180 + x_6)$

16. (a) **(3 points)** Find all values of the parameter t such that the vector $\begin{bmatrix} -1 \\ 0 \\ t \end{bmatrix}$ is in Span $\left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} -3 \\ -6 \\ 3 \end{bmatrix}, \begin{bmatrix} -5 \\ -13 \\ 2 \end{bmatrix} \right\}$

Solution:

$$\left[\begin{array}{ccc|c} 1 & -3 & -5 & -1 \\ 2 & -6 & -13 & 0 \\ -1 & 3 & 2 & t \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & -3 & -5 & -1 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & -3 & t-1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & -3 & -5 & -1 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & 0 & t-3 \end{array}\right] \Rightarrow \ t = 3.$$

(b) **(2 points)** Let $A = \begin{bmatrix} 1 & -\frac{3}{2} & -1 & \frac{1}{2} \\ 1 & -\frac{3}{2} & s & \frac{1}{2} \\ -2 & 3 & 2 & -1 \end{bmatrix}$. Find all values of s such that rank $A = 2$.

Solution:

$$\begin{bmatrix} 1 & -\frac{3}{2} & -1 & \frac{1}{2} \\ 1 & -\frac{3}{2} & s & \frac{1}{2} \\ -2 & 3 & 2 & -1 \end{bmatrix} \xrightarrow[R_3 \to R_3 + 2R_1]{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & -\frac{3}{2} & -1 & \frac{1}{2} \\ 0 & 0 & s+1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \ s \neq -1$$

17. In Belgium, the "goth fashion" clothing market is divided between two major retailers: *AngelNoir* and *GothCouture*. AngelNoir has 40% of the market, while GothCouture has the remaining 60% of the market. Market analysts predict that after one year, 30% of AngelNoir's customers will switch to GothCouture, while 10% of GothCouture's customers will switch to AngelNoir.

(a) **(1 point)** Write the migration matrix M.

Solution: $\begin{bmatrix} .7 & .1 \\ .3 & .9 \end{bmatrix}$ (Note that another possibility is $\begin{bmatrix} .9 & .3 \\ .1 & .7 \end{bmatrix}$

(b) **(2 points)** Determine the market share of each of the two retailers after a year according to the market analysts' prediction.

Solution: $\mathbf{x}_1 = M\mathbf{x}_0 = \begin{bmatrix} .7 & .1 \\ .3 & .9 \end{bmatrix}\begin{bmatrix} .4 \\ .6 \end{bmatrix} = \begin{bmatrix} .34 \\ .66 \end{bmatrix}$

Therefore the market shares are $\begin{cases} \text{AngelNoir:} & 34\% \\ \text{GothCouture:} & 66\% \end{cases}$

(c) **(3 points)** Assume that the retailers do not change their marketing strategies, and the customer migration trend continues for several years according to the market analysts' predictions. What will be the long term market share of the two retailers?

Solution:

$M\mathbf{x} = \mathbf{x} \Rightarrow (M - I_2)\mathbf{x} = \mathbf{0} \Rightarrow \begin{bmatrix} -.3 & .1 \\ .3 & -.1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\left[\begin{array}{cc|c} -.3 & .1 & 0 \\ .3 & -.1 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + R_1} \left[\begin{array}{cc|c} -.3 & .1 & 0 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to -\frac{10}{3}} \left[\begin{array}{cc|c} 1 & -\frac{1}{3} & 0 \\ 0 & 0 & 0 \end{array}\right] \Rightarrow \begin{cases} x_1 = \frac{1}{3}x_2 \\ x_1 + x_2 = 1 \end{cases}$

$\Rightarrow 1 - x_1 = x_2 = 3x_1 \Rightarrow x_1 = \frac{1}{4}$ and $x_2 = \frac{3}{4}$

Therefore the long term market shares are: $\begin{cases} \text{AngelNoir: } 25\% \\ \text{GothCouture: } 75\% \end{cases}$

18. (a) **(2 points)** Let $A = \begin{bmatrix} 4 & 0 & -2 \\ 1 & 2 & -3 \\ 1 & 0 & 1 \end{bmatrix}$. Write the characteristic polynomial of A and compute its eigenvalues.

Solution:
By expanding along the second column we have:

$$\begin{aligned} \det(A - \lambda I_3) &= \begin{vmatrix} 4-\lambda & 0 & -2 \\ 1 & 2-\lambda & -3 \\ 1 & 0 & 1-\lambda \end{vmatrix} = (2-\lambda) \begin{vmatrix} 4-\lambda & -2 \\ 1 & 1-\lambda \end{vmatrix} \\ &= -(\lambda-2)\left((4-\lambda)(1-\lambda) - (-2)\right) = -(\lambda-2)(\lambda^2 - 5\lambda + 6) \\ &= -(\lambda-2)(\lambda-2)(\lambda-3) \end{aligned}$$

and therefore the eigenvalues of A are $\lambda = 2$, $\lambda = 3$.

(b) **(4 points)** Let $B = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ -2 & -2 & 4 \end{bmatrix}$. Find a basis for the eigenspace of B that corresponds to the eigenvalue $\lambda = 2$.

Solution:

$$B - 2I_3 = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ -2 & -2 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ -2 & -2 & 2 \end{bmatrix} \xrightarrow[R_3 \to R_3 + 2R_1]{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \Rightarrow \quad x_1 = -x_2 + x_3$$

$$\Rightarrow \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_2 + x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{ basis for the eigenspace of } \lambda = 2 \ : \ \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

19. **(4 points)** Compute the inverse of $A = \begin{bmatrix} 1 & \frac{1}{3} & -1 \\ 0 & 2 & 1 \\ -1 & -\frac{1}{3} & 2 \end{bmatrix}$.

Solution:

$$\left[\begin{array}{ccc|ccc} 1 & \frac{1}{3} & -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ -1 & -\frac{1}{3} & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \to R_1 + R_3} \left[\begin{array}{ccc|ccc} 1 & \frac{1}{3} & -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[R_2 \to R_2 - R_3]{R_1 \to R_1 + R_3} \left[\begin{array}{ccc|ccc} 1 & \frac{1}{3} & 0 & 2 & 0 & 1 \\ 0 & 2 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to \frac{1}{2}R_2} \left[\begin{array}{ccc|ccc} 1 & \frac{1}{3} & 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - \frac{1}{3}R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{13}{6} & -\frac{1}{6} & \frac{7}{6} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array}\right]$$

Therefore $A^{-1} = \begin{bmatrix} \frac{13}{6} & -\frac{1}{6} & \frac{7}{6} \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 1 & 0 & 1 \end{bmatrix}$.

20. Let $A = \begin{bmatrix} 1 & \frac{1}{2} & 0 & 1 & -2 \\ 2 & 1 & 1 & 5 & -5 \\ -1 & -\frac{1}{2} & 1 & 2 & 1 \end{bmatrix}$.

(a) **(3 points)** Find a basis for $\operatorname{Col} A$.

Solution:

$$\begin{bmatrix} 1 & \frac{1}{2} & 0 & 1 & -2 \\ 2 & 1 & 1 & 5 & -5 \\ -1 & -\frac{1}{2} & 1 & 2 & 1 \end{bmatrix} \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & \frac{1}{2} & 0 & 1 & -2 \\ 0 & 0 & 1 & 3 & -1 \\ 0 & 0 & 1 & 3 & -1 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} 1 & \frac{1}{2} & 0 & 1 & -2 \\ 0 & 0 & 1 & 3 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Basis for $\operatorname{Col} A$: $\left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$.

(b) **(3 points)** Find a basis for $\operatorname{Nul} A$.

Solution: We use the reduced echelon form of part (a) to write $\begin{cases} x_1 = -\frac{1}{2}x_2 - x_4 + 2x_5 \\ x_2 : \text{free} \\ x_3 = -3x_4 + x_5 \\ x_4 : \text{free} \\ x_5 : \text{free} \end{cases}$

$$\Rightarrow \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}x_2 - x_4 + 2x_5 \\ x_2 \\ -3x_4 + x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ -3 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Basis for $\operatorname{Nul} A$: $\left\{ \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ -3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}$

21. An economy has two sectors: Energy and Service. We know that:

- In order to produce one unit of output, the Energy sector should consume $\frac{1}{5}$ units from itself and $\frac{1}{5}$ units from the Service sector.
- In order to produce one unit of output, the Service sector should consume $\frac{3}{4}$ units from itself and $\frac{1}{4}$ units from the Energy sector.

(a) **(1 point)** Write down the consumption matrix for this economy.

Solution: $C = \begin{bmatrix} \frac{1}{5} & \frac{1}{4} \\ \frac{1}{5} & \frac{3}{4} \end{bmatrix}$ (Note that $C = \begin{bmatrix} \frac{3}{4} & \frac{1}{5} \\ \frac{1}{4} & \frac{1}{5} \end{bmatrix}$ is another valid answer.)

(b) **(3 points)** Calculate the production levels that are required to meet a final demand of 9 units from the Energy sector and 12 units from the Service sector.

Solution: $I_2 - C = \begin{bmatrix} \frac{4}{5} & -\frac{1}{4} \\ -\frac{1}{5} & \frac{1}{4} \end{bmatrix} \Rightarrow (I_2 - C)^{-1} = \frac{1}{(\frac{3}{20})} \begin{bmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{1}{5} & \frac{4}{5} \end{bmatrix} = \begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{4}{3} & \frac{16}{3} \end{bmatrix}$

$\Rightarrow \mathbf{x} = \begin{bmatrix} \frac{5}{3} & \frac{5}{3} \\ \frac{4}{3} & \frac{16}{3} \end{bmatrix} \begin{bmatrix} 9 \\ 12 \end{bmatrix} = \begin{bmatrix} 35 \\ 76 \end{bmatrix}$

and production levels are $\begin{cases} \text{Energy:} & 35 \\ \text{Private:} & 76. \end{cases}$